

Lecture 9: Trading with Price Impact, and the TWAP Strategy

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9.1 Market Mechanics & Price Impact

In high-frequency and algorithmic trading, liquidating a large position of $\mathfrak{N}$ shares over a finite horizon $[0, T]$ cannot be executed instantaneously without causing severe adverse price movement. We parameterize the portfolio inventory at time $t \in [0, T]$ by Q_t , where the trading speed ν_t represents the continuous rate of selling,

$$dQ_t = -\nu_t dt .$$

Executing trades impacts market prices through two distinct physical mechanisms governed by impact functions $g(\nu)$ and $f(\nu)$.

Permanent Price Impact $g(\nu)$: Large sell orders continuously absorb liquidity across the order book, persistently shifting the mid-price S_t . Under a general non-decreasing permanent impact function $g(\nu)$ with $g(0) = 0$, the unaffected mid-price follows

$$dS_t = -g(\nu_t)dt + \sigma dW_t ,$$

where W_t is a standard Brownian motion and $\sigma > 0$ denotes asset volatility. This is a **reduced-form permanent price impact**, which could be given a macroeconomic interpretation if the model is embedded within a larger equilibrium framework.

Temporary Price Impact $f(\nu)$: To execute at speed ν_t , the trader must walk down the limit order book, incurring instantaneous execution slippage beyond the mid-price S_t . Under a temporary impact function $f(\nu)$ with $f(0) = 0$ and $f'(\nu) > 0$, the effective per-share execution price $\hat{S}_t$ realized by the trader is

$$\hat{S}_t = S_t - f(\nu_t) .$$

Linear Temporary Impact Specification: In the baseline setting, we model temporary price impact linearly with parameter $k > 0$, giving the execution price

$$\hat{S}_t = S_t - k\nu_t .$$

9.2 Stochastic Control Problem Formulation

Selling $dQ_t = -\nu_t dt$ shares yields continuous cash inflows at rate $\hat{S}_t \nu_t = (S_t - f(\nu_t)) \nu_t = S_t \nu_t - L(\nu_t)$, where $L(\nu_t) \equiv f(\nu_t) \nu_t$ represents the continuous execution cost rate incurred from trading at speed ν_t .

In a general stochastic control framework, an agent liquidating $Q_0 = q$ shares maximizes expected net revenues subject to speed penalty $L(\nu_s)$ and terminal liquidation penalty $\Psi(Q_T)$,

$$J(\nu) = \mathbb{E} \left[\int_0^T (S_u \nu_u - L(\nu_u)) du - \Psi(Q_T) \right] .$$

Defining the value function

$$H(t, s, q) = \sup_{\nu \in \mathcal{A}} \mathbb{E}_{t,s,q} \left[\int_t^T (S_u \nu_u - L(\nu_u)) du - \Psi(Q_T) \right] ,$$

dynamic programming yields the general HJB equation,

$$\frac{\partial H}{\partial t} + \frac{1}{2} \sigma^2 \frac{\partial^2 H}{\partial s^2} + \max_{\nu} \left\{ -\nu \frac{\partial H}{\partial q} - g(\nu) q \frac{\partial H}{\partial s} + s\nu - L(\nu) \right\} = 0 .$$

9.3 Linear-Quadratic with No Permanent Impact

We specialize the general setup to linear temporary impact $f(\nu) = k\nu$, giving running cost $L(\nu) = k\nu^2$, zero permanent impact ($g(\nu) = 0$) so that

$$dS_t = \sigma dW_t ,$$

and a hard requirement of complete liquidation ($Q_T = 0$).

The agent minimizes the total execution cost EC^ν , defined as initial paper value $\mathfrak{N}S_0$ minus realized revenue,

$$EC^\nu = \mathfrak{N}S_0 - \mathbb{E} \left[\int_0^T \hat{S}_t \nu_t dt \right] .$$

Equivalently, the agent maximizes expected revenues over $[t, T]$ for $Q_t = q$ and $S_t = s$,

$$H(t, s, q) = \sup_{\nu \in \mathcal{A}} \mathbb{E}_{t,s,q} \left[\int_t^T (S_u - k\nu_u) \nu_u du \right] , \tag{9.1}$$

subject to $dQ_t = -\nu_t dt$, $Q_t = q$, and $Q_T = 0$.

9.3.1 Solving for the Optimal

Step 1: The HJB Equation Substituting $L(\nu) = k\nu^2$ into the general HJB equation yields

$$\frac{\partial H}{\partial t} + \frac{1}{2} \sigma^2 \frac{\partial^2 H}{\partial s^2} + \max_{\nu} \left\{ -\nu \frac{\partial H}{\partial q} + s\nu - k\nu^2 \right\} = 0 .$$

Step 2: Value Function Decomposition Because mid-price S_t is an unbiased martingale ($\mathbb{E}[dS_t] = 0$), holding q shares has mark-to-market value qs . We propose the additive ansatz

$$H(t, s, q) = qs + h(t, q) .$$

Calculating partial derivatives $\frac{\partial H}{\partial t} = \frac{\partial h}{\partial t}$, $\frac{\partial H}{\partial s} = q$, $\frac{\partial^2 H}{\partial s^2} = 0$, and $\frac{\partial H}{\partial q} = s + \frac{\partial h}{\partial q}$, substitution yields

$$\frac{\partial h}{\partial t} + \max_{\nu} \left\{ -\nu \left(s + \frac{\partial h}{\partial q} \right) + s\nu - k\nu^2 \right\} = 0 ,$$

which reduces to

$$\frac{\partial h}{\partial t} + \max_{\nu} \left\{ -k\nu^2 - \nu \frac{\partial h}{\partial q} \right\} = 0 . \quad (9.2)$$

Step 3: First-Order Condition (FOC) Differentiating inside the maximum with respect to ν gives

$$-2k\nu - \frac{\partial h}{\partial q} = 0 \implies \nu^*(t, q) = -\frac{1}{2k} \frac{\partial h}{\partial q} .$$

Substituting ν^* back into equation (9.2) yields

$$\frac{\partial h}{\partial t} + \frac{1}{4k} \left(\frac{\partial h}{\partial q} \right)^2 = 0 .$$

Step 4: Quadratic Value Function Ansatz Proposing $h(t, q) = -h_2(t)q^2$, we have $\frac{\partial h}{\partial t} = -\frac{d}{dt}h_2(t)q^2$ and $\frac{\partial h}{\partial q} = -2h_2(t)q$. The PDE transforms to

$$-\frac{d}{dt}h_2(t)q^2 + \frac{(-2h_2(t)q)^2}{4k} = 0 \implies -\frac{d}{dt}h_2(t) + \frac{h_2^2(t)}{k} = 0 .$$

Separating variables yields

$$\frac{dh_2}{h_2^2} = \frac{1}{k} dt \implies -\frac{1}{h_2(t)} = \frac{t - C}{k} \implies h_2(t) = \frac{k}{C - t} .$$

Enforcing $Q_T = 0$ requires terminal execution costs to explode as $t \rightarrow T$, fixing $C = T$,

$$h_2(t) = \frac{k}{T - t} .$$

Step 5: Optimal Policy and Trajectory The optimal feedback trading speed is

$$\nu^*(t, q) = \frac{2h_2(t)q}{2k} = \frac{q}{T - t} .$$

Substituting ν^* into inventory dynamics $\frac{d}{dt}Q_t^* = -\nu^*(t, Q_t^*)$ gives

$$\frac{d}{dt}Q_t^* = -\frac{Q_t^*}{T - t} .$$

Integrating from 0 to t with $Q_0^* = \mathfrak{N}$ yields

$$Q_t^* = \mathfrak{N} \left(1 - \frac{t}{T} \right) \quad \text{and} \quad \nu_t^* = -\frac{d}{dt}Q_t^* = \frac{\mathfrak{N}}{T} . \quad (9.3)$$

9.3.2 Economic Interpretation

Optimality of Time-Weighted Average Price (TWAP): For a risk-neutral agent, the optimal policy liquidates inventory at a strictly constant rate $\nu_t^* = \frac{\mathfrak{N}}{T}$, proving TWAP is optimal under linear temporary impact.

Expected Execution Cost via Jensen's Inequality: The total execution cost EC^* evaluates to

$$EC^* = \mathbb{E} \left[\int_0^T k(\nu_t^*)^2 dt \right] = k \int_0^T \left(\frac{\mathfrak{N}}{T} \right)^2 dt = \frac{k\mathfrak{N}^2}{T} .$$

By Jensen's inequality, any non-constant speed ν_t with $\int_0^T \nu_t dt = \mathfrak{N}$ incurs strictly higher cost,

$$\int_0^T \nu_t^2 dt \geq \frac{1}{T} \left(\int_0^T \nu_t dt \right)^2 = \frac{\mathfrak{N}^2}{T} .$$

Absence of Market Timing: Because mid-price S_t has zero drift ($\mathbb{E}[dS_t] = 0$), a risk-neutral agent gains no expected benefit from attempting to time price fluctuations.

9.4 Remarks

1. The optimal liquidation problem presented here is originally credited to [1].
2. This lecture was based on Chapter 6.1, 6.2 and 6.3 in [2]

References

- [1] R. Almgren and N. Chriss. Optimal execution of portfolio transactions. *Journal of Risk*, 3:5–40, 2001.
- [2] Á. Cartea, S. Jaimungal, and J. Penalva. *Algorithmic and high-frequency trading*. Cambridge University Press, 2015.